Impulse changes momentum

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Learning target: Connect force over time to momentum change.

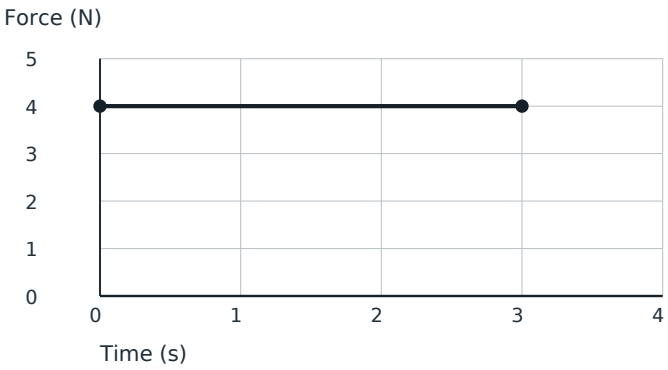
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 16.*
g=3 N/kg; force 21 N inward.

02. Worked example: explain the model
Area 12 N·s; N·s=(kg·m/s²) s=kg·m/s.

03. Read and interpret the model
4 N×3 s=12 N·s impulse in positive direction.

Model / diagram check



04. Apply the physics
3 m/s right.

05. Explain your reasoning
Impulse/momentum change equal. Peak force and duration need not be equal.

06. Review: connect an earlier idea *Review: Unit 2, Day 7.*
4 N/kg toward the center; divide the surface value by 36.

07. Exit check
10 N·s right; 5 m/s right.

08. STRETCH: test the model
Examples 3 N for 4 s and 12 N for 1 s; equal areas with different heights and widths.

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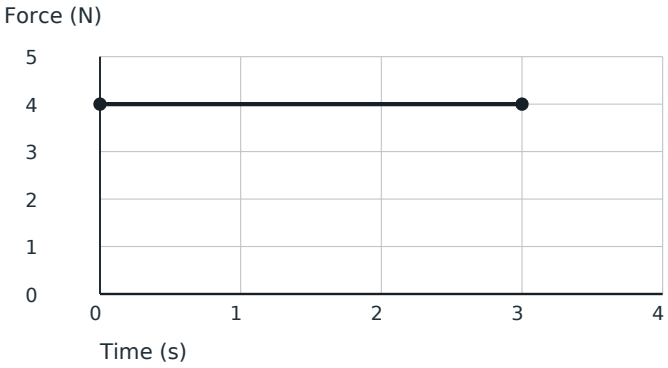
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 16.*
v=5 m/s; a≈2.778 m/s²; T≈11.31 s. Speed constant, direction changing.

02. Worked example: explain the model
Area 12 N·s; N·s=(kg·m/s²) s=kg·m/s.

03. Read and interpret the model
4 N×3 s=12 N·s impulse in positive direction.

Model / diagram check



04. Apply the physics
J=0.5(-14)=-7 N·s; F=-350 N.

05. Explain your reasoning
Impulse/momentum change equal. Peak force and duration need not be equal.

06. Review: connect an earlier idea *Review: Unit 2, Day 7.*
Center distances 4 R and 6 R; altitudes h/R=3 and 5.

07. Exit check
J=0.5(0.20)(30)=3 N·s; Δv=5 m/s in force direction.

08. STRETCH: test the model
Impulse 2(-2-3)=-10 N·s; average force -100 N.